EE2002-Tutorial 6 Solutions

1.A set-dominate latch has a set and reset input. It differs from a conventional SR latch in that an attempt to simultaneously set and reset results in setting the latch. The circuit for this is shown in Figure 1. Obtain the truth table (next-state table) for this latch.

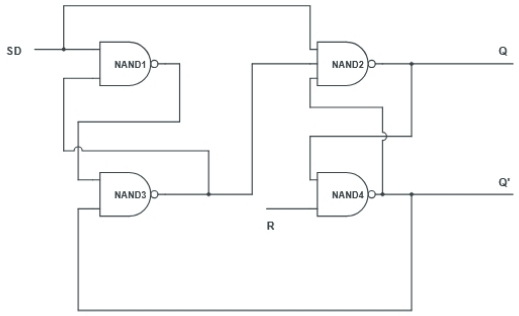


Figure 1: Set-dominate Latch

Solution:

$Q(t)$	SD	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

From table:

$$Q(t+1) = \sum(2,3,4,6,7) = SD + R'Q(t)$$

2.Consider the T flip-flop in Figure 3. Choose appropriate waveforms for the Clock signal and the T input and draw a timing diagram. Show Q as well as $\overline{Q}$.

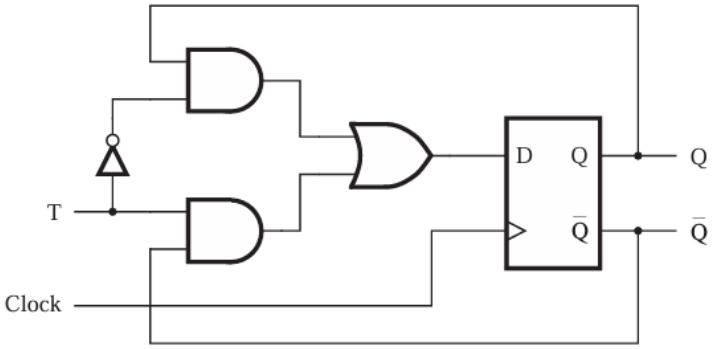


Figure 2: T flip-flop

Solution:

The D flip-flop is a versatile storage element that can be modified using simple logic. An interesting modification, uses a positive-edge-triggered D flip-flop with feedback connections. The input D is set to either Q or $\overline{Q}$ based on the control signal T.

On each positive clock edge: If $T = 0$, then $D = Q$, and the state remains unchanged, i.e., $Q(t+1) = Q(t)$.

If $T = 1$, then $D = \overline{Q}$, causing the state to toggle, i.e., $Q(t+1) = \overline{Q(t)}$.

This operation is summarized in the characteristic table :

T	$Q(t)$	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

T	$Q(t+1)$
0	$Q(t)$
1	$\overline{Q(t)}$

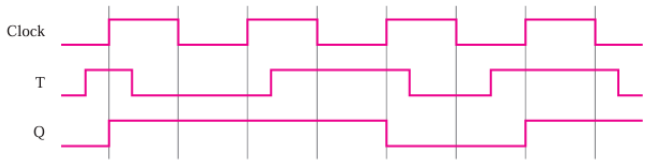


Figure 3: T flip-flop timing diagram

Q' waveform is complement to Q waveform.

3.Consider the JK flip-flop shown in Figure 4. Write the characteristic equation and the characteristic table for this circuit.

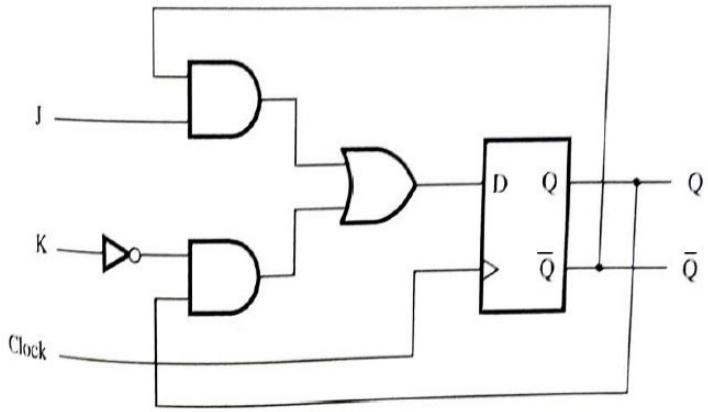


Figure 4: JK Flip-Flop

Solution:

The given JK flip-flop circuit is implemented using a D flip-flop. The input to the D flip-flop is given by:

$$D = J\bar{Q} + \bar{K}Q$$

Since a D flip-flop follows the rule $Q^+ = D$, the next-state equation becomes:

$$Q^+ = J\bar{Q} + \bar{K}Q$$

This is the characteristic equation of the JK flip-flop.

The characteristic table is constructed based on all possible values of J, K, Q and the corresponding next state Q^+:

J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

From the table:

$$Q^+ = \sum(1, 4, 5, 6) = J\bar{Q} + \bar{K}Q$$

J	K	Q^+
0	0	Q
0	1	0
1	0	1
1	1	$\bar{Q}$

4.A four-bit shift register, which shifts data right ($Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3$) is shown in Figure 5. It is provided an initial state of $Q_0Q_1Q_2Q_3 = 1000$.

(a) Write the 4-bit values for $Q_0, Q_1, Q_2,$ and Q_3 after each clock pulse until the pattern 1000 reappears.

(b) To what values should the shift register be initialized so that the pattern 1000 occurs after 2 clock pulses?

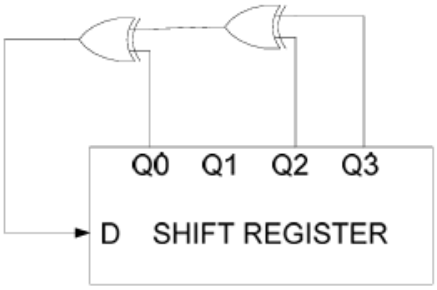


Figure 5: 4-bit Shift Register

Solution:

The next state of Q_0 is determined by:

$$D_{\text{next}} = Q_0 \oplus Q_2 \oplus Q_3$$

(a) Step-by-Step Shift Operations

The following table shows the values of Q_0, Q_1, Q_2, Q_3 after each clock pulse:

Clock Pulse	Q_0	Q_1	Q_2	Q_3	$D_{\text{next}} = Q_0 \oplus Q_2 \oplus Q_3$
Initial (0)	1	0	0	0	$1 \oplus 0 \oplus 0 = 1$
1st Pulse	1	1	0	0	$1 \oplus 0 \oplus 0 = 1$
2nd Pulse	1	1	1	0	$1 \oplus 1 \oplus 0 = 0$
3rd Pulse	0	1	1	1	$0 \oplus 1 \oplus 1 = 0$
4th Pulse	0	0	1	1	$0 \oplus 1 \oplus 1 = 0$
5th Pulse	0	0	0	1	$0 \oplus 0 \oplus 1 = 1$
6th Pulse	1	0	0	0	$1 \oplus 0 \oplus 0 = 1$

Since the pattern 1000 reappears at the 6th clock pulse, the cycle length is 6.

(b) Initial Values for 1000 to Appear After 2 Cycles

We need to find initial values so that the register holds 1000 after two clock pulses.

Observing the table, we see that 1000 appears at clock pulses 6 and 0. Looking back two steps before 1000, we find:

Clock Pulse	Q_0	Q_1	Q_2	Q_3
$6thPulse$	1	0	0	0
$5thPulse$	0	0	0	1
$4thPulse$	0	0	1	1

Thus, to get 1000 after 2 clock pulses, we should initialize the register as:

$$Q_0 = 0, \quad Q_1 = 0, \quad Q_2 = 1, \quad Q_3 = 1$$

5.Construct the truth table (next state table) of the sequential circuit in Figure 6 and comment on its operation.

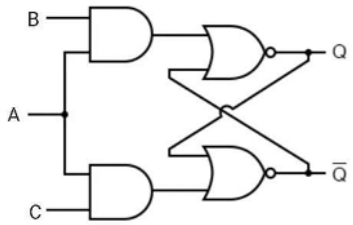


Figure 6: Sequential Circuit

Solution:

The given circuit consists of two AND gates and two NOR gates, forming an SR latch. Let us analyze its behavior step by step.

Using Boolean algebra, the circuit can be analyzed as follows:

$$Q = \overline{(A \cdot B) + \bar{Q}}$$
$$\bar{Q} = \overline{(A \cdot C) + Q}$$

This shows that the circuit behaves like an SR latch with inputs $A \cdot B$ and $A \cdot C$.

The circuit operates as follows:

- If $A = 0$, both inputs to the NOR gates become 0, causing the latch to maintain its previous state.
- If $A = 1$, the circuit behaves based on the values of B and C:
  - If $B = 1$ and $C = 0$, the output Q is Reset to 0.
  - If $B = 0$ and $C = 1$, the output Q is Set to 1.
  - If $B = 1$ and $C = 1$, it results in an undefined state.

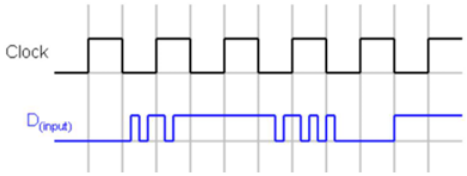


Figure 7: Timing Diagram

The following table represents the behavior of the given sequential circuit:

A	B	C	Q^+	Q^+	State
0	X	X	Q	$\bar{Q}$	Previous state
1	0	0	Q	$\bar{Q}$	Previous state
1	0	1	1	0	set
1	1	0	0	1	Reset
1	1	1	0	0	Invalid

The given circuit functions as a gated SR latch.

6.Trace the outputs of a D latch and D flip-flop (positive-edge-triggered) for the input and clock signals provided in Figure 7. Compare the outputs and comment on the observations.

Solution:

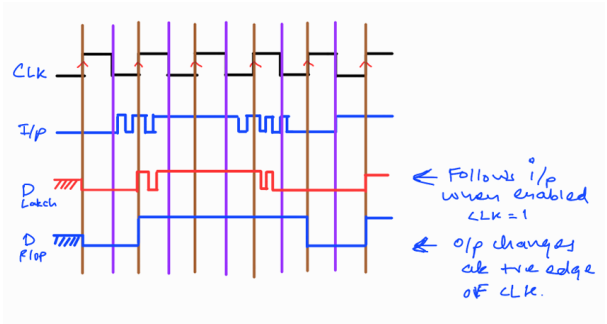


Figure 8: Analysis of Waveform

7.The initial content of a four-bit register is 0110. The register is shifted six times to the right with the serial input being 1011100. Assume the rightmost bit is the LSB. What is the content of the register after each shift.

Solution:

Each right shift operation moves the bits one position to the right, inserting the next bit from the serial input into the leftmost position (MSB). The rightmost bit (LSB) is shifted out in each step.

The initial content of the register is 0110. In each shift, the rightmost bit is discarded, and the next bit from the serial input is inserted at the leftmost position. The final content of the register is 1011.

Shift No.	Serial Input	Register Content
Initial	-	0110
1	0	0011
2	0	0001
3	1	1000
4	1	1100
5	1	1110
6	0	0111
7	1	1011

8.A 74178 shift register is described by Figure 9. All state changes occur on the 1-0 transition of the clock. The shift register is connected as shown in Figure 9. Complete the timing diagram.

Solution:

The shift register follows the behavior described below:

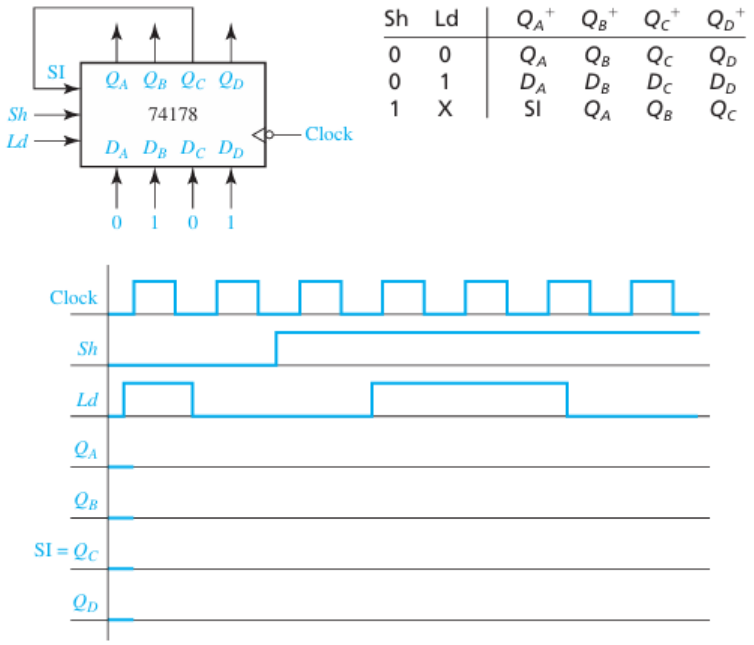
- When `Sh` = 0 and `Ld` = 0, the register retains its previous state.
- When `Sh` = 0 and `Ld` = 1, parallel data is loaded into the register.
- When `Sh` = 1, the data shifts right, and the serial input (`SI`) is loaded into Q_A.
- If both `Sh` and `Ld` are 1, shifting takes precedence over loading.

The timing diagram illustrates the behavior of the shift register based on the control signals and clock transitions.

Observations: Data is loaded into the register when `Ld` = 1 and `Sh` = 0. When `Sh` = 1, data shifts right on each negative clock edge.

The serial input (`SI`) replaces Q_A during shifting. If `Sh` and `Ld` are both 1, shifting takes priority.

This confirms the expected operation of the 74178 shift register.



Sh	Ld	Q_A^+	Q_B^+	Q_C^+	Q_D^+
0	0	Q_A	Q_B	Q_C	Q_D
0	1	D_A	D_B	D_C	D_D
1	X	SI	Q_A	Q_B	Q_C

Figure 9: 74178 Shift register

Notice that *Sh* overrides *Ld* when *Sh* = *Ld* = 1

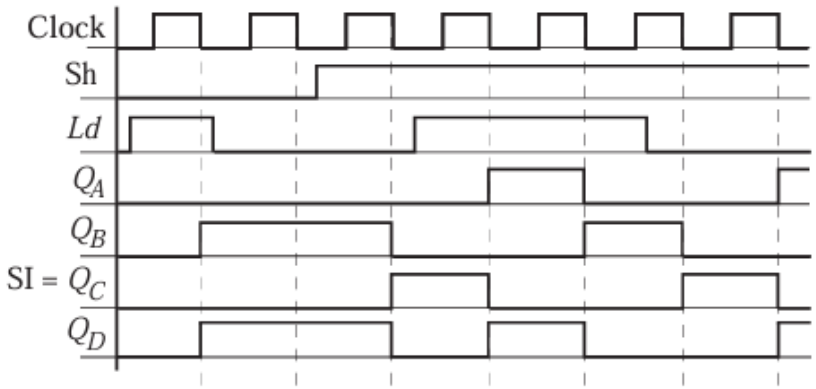


Figure 10: Timing Diagram of the 74178 Shift Register